SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

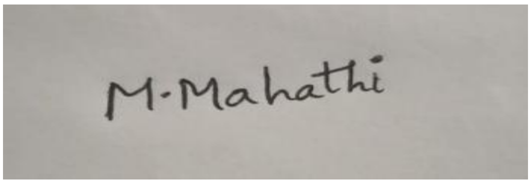
Annexure B
Analytical Problem Solving

Criteria	Problem Understanding ***(2)***	Method Selection ***(2)***	Solution Process ***(2.5)***	Accuracy of Calculation ***(2)***	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I __M.Mahathi(192324098)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

25|7|27

ACTIVITY – 04

1) Suppose that the data for analysis include the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

(a) What is mean of data?

(b) what is mode of data? Comment on data's modal

(c) what is midrange of the data?

(d) can you find (roughly) the first Quartile (Q1) and third quartile (Q3) of the data?

(e) Give the five-number summary of the data

(f) Show a boxplot of the data.

(a) Mean ⇒ $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$

$$= \frac{(13+15+16+16+19+20+20+21+22+22+25+25+25+25+30+33+33+35+35+35+35+36+40+46+52+70)}{26}$$

$$= \frac{\cancel{764}}{\cancel{26}} = \frac{382}{\cancel{13}} = \frac{809}{27} =$$

$$= 29.96$$

(b) Median ⇒

$$\frac{n+1}{2} = \frac{27+1}{2} = \frac{28}{2} = 14 \text{ th position}$$

Median = 25

(b) Mode :

25 and 35 are coming continuously

As only 2 values are repeating, we have to call them by bimodal

(c) Mid range → $\frac{\text{Maxvalue} + \text{Minvalue}}{2}$

$\Rightarrow \frac{70+13}{2} = \frac{83}{2} = 41.5$

(d) First Quartile (Q1)

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25

Median(Q_1) $\Rightarrow \frac{13+1}{2} = \frac{14}{2} = 7$th value

∴ Median(Q1) = 20

Third Quartile (Q3)

30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

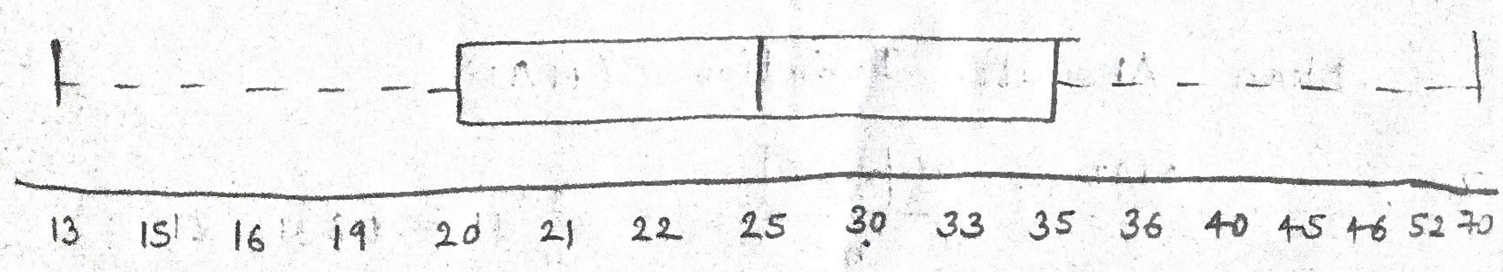
Median (Q3) → $\frac{13+1}{2} = \frac{14}{2} = 7$th value

∴ Median (Q3) = 35

(e) Five number Summary : min, Q1, median, Q3, max

Min = 13, Q1 = 20, median = 25, Q3 = 35, max = 70

(f) Box Plot



2) Given the following measurements for variable age.

18, 22, 25, 42, 28, 43, 33, 35, 56, 28, standardize the variable by following:

computing the mean absolute deviation of age.

Mean $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$

$$= \frac{18+22+25+42+28+43+33+35+56+28}{10}$$

$$= \frac{330}{10} = 33$$

x(age)	$\lvert x-\bar{x} \rvert$
18	$\lvert 18-33 \rvert = 15$
22	$\lvert 22-33 \rvert = 11$
25	$\lvert 25-33 \rvert = 8$
42	$\lvert 42-33 \rvert = 9$
28	$\lvert 28-33 \rvert = 5$
43	$\lvert 43-33 \rvert = 10$
33	$\lvert 33-33 \rvert = 0$
35	$\lvert 35-33 \rvert = 2$
56	$\lvert 56-33 \rvert = 23$
28	$\lvert 28-33 \rvert = 5$

Sum of all absolute deviation $\Rightarrow$

$15+11+8+9+5+10+0+2+23+5$

$= 88$

Mean Absolute Deviation (MAD)

$$MAD = \frac{\sum \lvert x-\bar{x} \rvert}{n} = \frac{88}{10} = 8.8$$

3) Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means to smooth the above data, using a bin depth of 3. Illustrate your steps.

Number of values = 27

Bin depth = 3

$$\text{Count of Bin} = \frac{\text{Number of values (occurence)}}{\text{Bin depth}} = \frac{27}{3} = 9$$

Bin	Values	Mean
B1	13, 15, 16	(13+15+16)/3 = 14.67
B2	16, 19, 20	(16+19+20)/3 = 18.33
B3	20, 21, 22	(20+21+22)/3 = 21.00
B4	22, 25, 25	(22+25+25)/3 = 24.00
B5	25, 25, 30	(25+25+30)/3 = 26.67
B6	33, 33, 35	(33+33+35)/3 = 33.67
B7	35, 35, 35	(35+35+35)/3 = 35.00
B8	36, 40, 45	(36+40+45)/3 = 40.33
B9	46, 52, 70	(46+52+70)/3 = 56.00

Replace the mean for all 3 values in the Bin

Bin	Original Values	Bin Mean	Smoothed Values
B1	13,15,16	14.67	14.67, 14.67, 14.67
B2	16,19,20	18.33	18.33, 18.33, 18.33
B3	20,21,22	21.00	21.00, 21.00, 21.00
B4	22,25,25	24.00	24.00, 24.00, 24.00
B5	25,25,30	26.67	26.67, 26.67, 26.67
B6	33,33,35	33.67	33.67, 33.67, 33.67
B7	35,35,35	35.00	35.00, 35.00, 35.00
B8	36,40,45	40.33	40.33, 40.33, 40.33
B9	46,52,70	56.00	56.00, 56.00, 56.00

Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000

(a) min-max normalization by setting min=0 and max = 1.

(b) z-score normalization.

(a). Min-Max Normalization

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Max = 1000

Min = 200

$$x' = \frac{x - 200}{1000 - 200} = \frac{x - 200}{800}$$

Original value	Calculation	Normalized value
200	(200−200)/800	0
300	(300−200)/800	0.125
400	(400−200)/800	0.25
600	(600−200)/800	0.5
1000	(1000−200)/800	1

(b) Z-Score Normalization.

$$Z = \frac{x-\mu}{\sigma}$$

i) Mean, μ $= \dfrac{200+300+400+600+1000}{5} = \dfrac{2500}{5} = 500$

ii) Standard Deviation

x	$x-\mu$	$(x-\mu)^2$
200	200 − 500 = −300	$(-300)^2 = 90000$
300	300 − 500 = −200	$(-200)^2 = 40000$
400	400 − 500 = −100	$(-100)^2 = 10000$
600	600 − 500 = 100	$(100)^2 = 10000$
1000	1000 − 500 = 500	$(500)^2 = 250000$

$\Sigma(x-\mu)^2 = 90000 + 40000 + 10000 + 10000 + 250000$

$= 400000$

Variance, $\sigma^2 = \dfrac{400000}{5} = 80000$

Standard Deviation $= \sigma = \sqrt{\dfrac{\Sigma(x-\mu)^2}{n}}$

$\sigma = \sqrt{80000} = 282.84$

Compute z-score

Original Value	Calculation	z-score
200	(200-500)/282.84	-1.06
300	(300-500)/282.84	-0.71
400	(400-500)/282.84	-0.35
600	(600-500)/282.84	0.35
1000	(1000-500)/282.84	1.77

∴ z-scores are -1.06, -0.71, -0.35, 0.35, 1.77

5) Use the two methods below to normalize the following group of data: 200.

5) Suppose a group of 12 sales price records:
5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215. Partition them into three bins by each of the following methods.

(a) equal-frequency partitioning.

(b) equal-width partitioning

(c) clustering

Number of records = 12

Number of bins = 3

Records per bin = $\frac{12}{3}$ = 4 values

Bin	Values
B1	5, 10, 11, 13
B2	15, 35, 50, 55
B3	72, 92, 204, 215

(b) width = $\dfrac{\text{Maximum} - \text{Minimum}}{\text{Number of bins}}$

$= \dfrac{215 - 5}{3} = \dfrac{210}{3} = 70$

Intervals

Bin 1 - 5-74

Bin 2 - 75-144

Bin 3 - 145-215

Bin	Interval	Values
B1	5-74	5, 10, 11, 13, 15, 35, 50, 55, 72
B2	75-144	92
B3	145-215	204, 215

(c) clustering (similar values together)

Cluster	Values
Cluster 1	5, 10, 11, 13, 15
Cluster 2	35, 50, 55, 72, 92
Cluster 3	204, 215

25/7/26

6) Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following result.

age	23	23	27	27	39	41	47	49	50
% fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

age	52	54	54	56	57	58	58	60	61
% fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median and standard deviation of age and % fat

(b) Draw the boxplots for age and % fat

(c) Draw a scatter plot and a q-q plot based on z-score normalization. 2 variables

(d) Normalize the two variables based on z-score normalization.

(e) Calculate the correlation coefficient (Person's product moment coefficient). Are these two variables positevely or negatively correlated?

(a) Mean, Median and Standard Deviation

	Age	% fat
Mean	46.44	28.78
Median	51.00	30.70
Standard Deviation	12.85	8.99

Five-Number Summary:

Minimum = 23

Q1 = 33

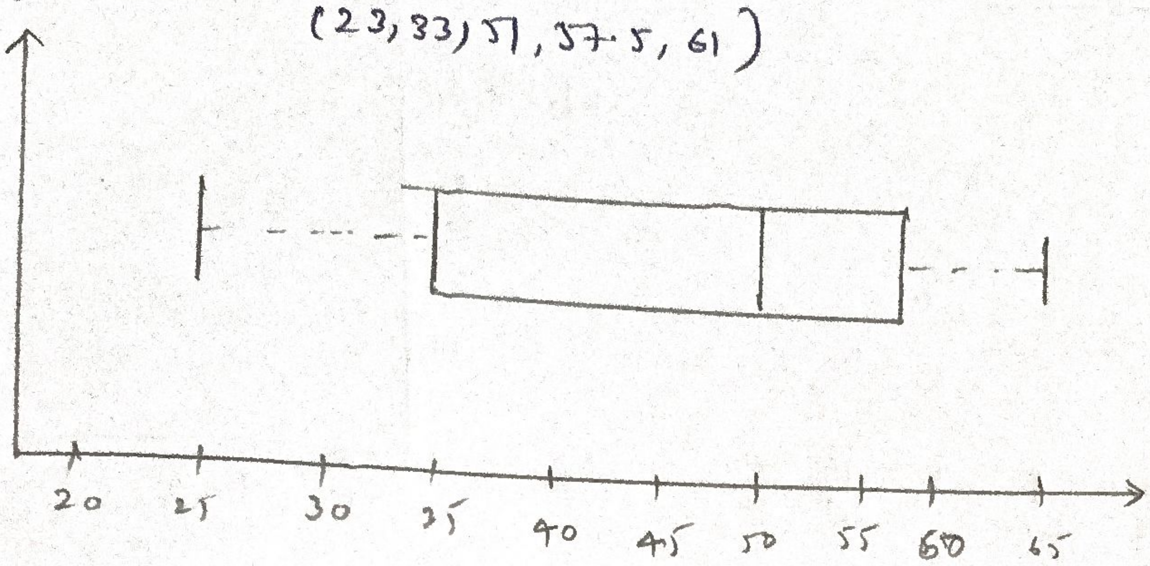
Maximum = 61

Median = 51

Q2 = 57

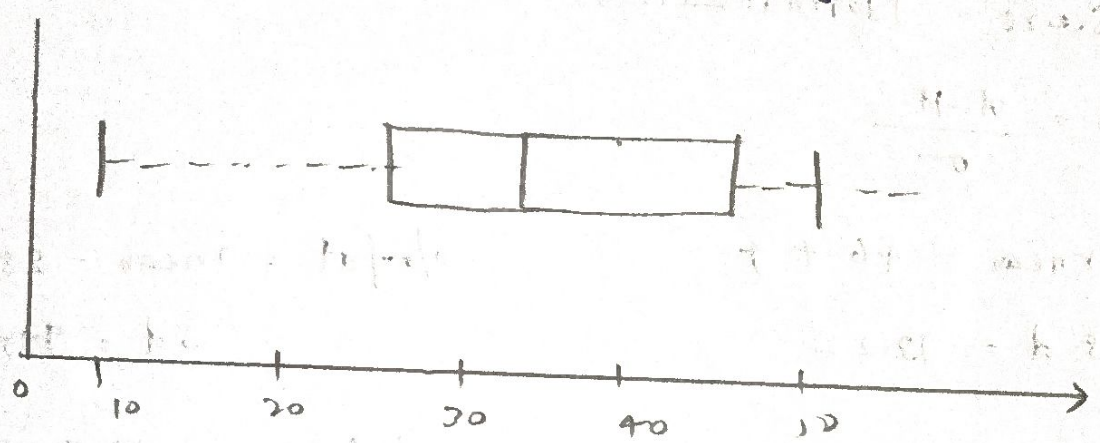
(b) Box plot :-

(23, 33, 51, 57.5, 61)



% fat Box pot

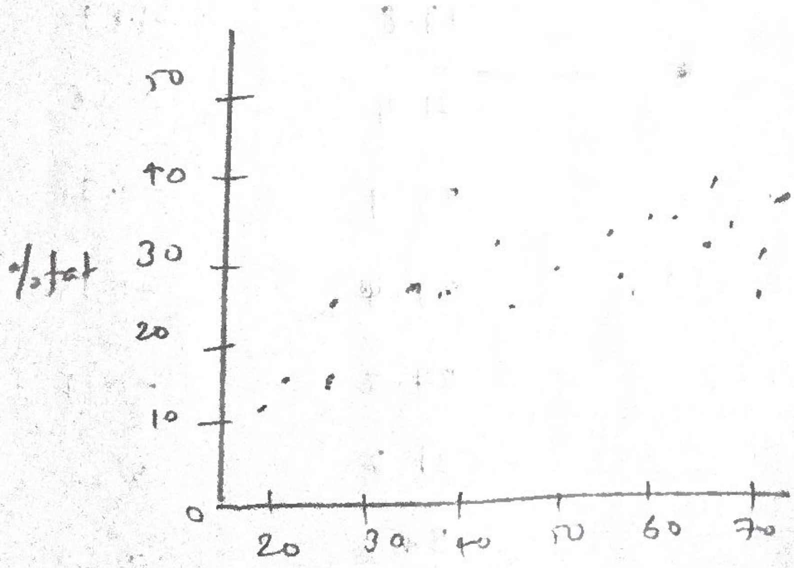
(7.8, 26.2, 30.7, 34.35, 42.5)



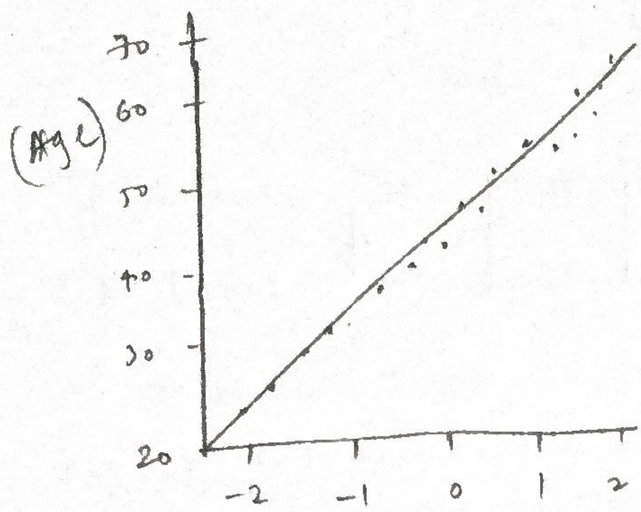
(c) Scatter pot

x_axis : Age

y- axis: % fat

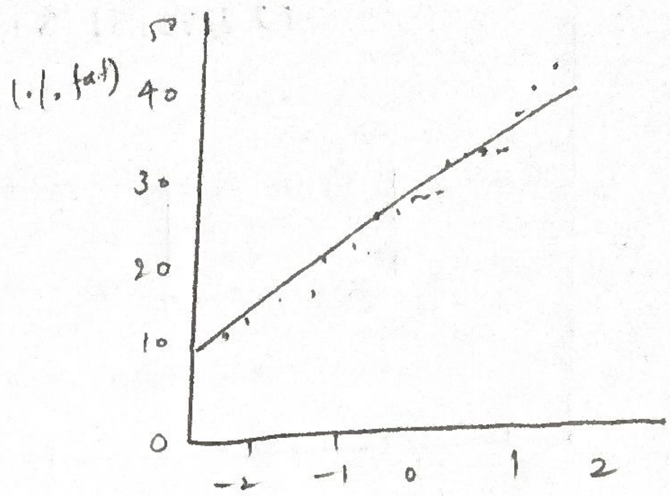


Q-Q plot : Age



Theoretical Quantiles

Q-Q plot : % fat.



Theoretical Quantiles

(D) Z-Score Normalization

$$z = \frac{x - \mu}{\sigma}$$

Age: mean = 46.44

s.d = 13.22

% fat = mean = 28.78

sd = 9.15

Age	Z-Score	% fat	Z-score
23	-1.77	9.5	-2.08
23	-1.77	26.5	-0.25
27	-1.47	7.8	-2.27
27	-1.47	17.8	-1.19
39	-0.56	31.4	0.28
41	-0.41	25.9	-0.31
47	+0.04	27.4	-0.15
49	0.19	27.2	-0.17
50	0.42	31.2	0.26
52	0.57	34.6	0.63
54	0.57	42.5	1.48
54	0.72	28.8	0.000
56	0.80	33.4	0.50
57	0.87	30.2	0.15
58	0.87	34.1	0.57
58	1.03	32.9	0.45
60	1.03	41.2	1.34
[illegible]	1.10	[illegible]	[illegible]

$$r = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\sqrt{\Sigma (x-\bar{x})^2 \, \Sigma (y-\bar{y})^2}}$$

x = Age , y = % fat

$\bar{x} = 46.44$, $\bar{y} = 28.78$

$r = 0.818$

$\therefore r > 0$

$\therefore$ Correlation is positive

$\therefore$ Age and % Fat are strongly positively correlate.